

Hydraulic Properties of Various Weathered Granitic Bedrock in Headwater Catchments

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Recent studies have emphasized the importance of bedrock in hydrologic processes occurring in headwater catchments. To understand water flow processes through variously weathered bedrock, we measured the saturated hydraulic conductivity, K_s , and water retention characteristics of weakly to highly weathered Tanakami granite and Rokko granite core samples. On the basis of these core-scale properties, along with the core shape and in situ K_s measurements, we defined two groups of bedrock: C_M class (weakly weathered) and C_L to D_L class (moderately to highly weathered). The C_M class bedrock cores had almost no effective porosity (i.e., the amount of porosity that effectively contributes to water flow) and therefore extremely small core-scale K_s , indicating that the matrix could be regarded as essentially impermeable. The in situ K_s was much larger than the core-scale values, however, and the core shape showed apparent fractures, suggesting that water did flow preferentially through the fractures. The volumetric water content of the C_L –to– D_L –class bedrock water retention curves changed little in the dry range but changed gradually in the wet range, resulting in a moderate core-scale K_s of 10^{-5} to 10^{-3} cm s $^{-1}$. The core-scale K_s values were well explained by the parameters characterizing the water retention curve. The similarity of the in situ K_s to the core-scale values, and the lack of fractures in the core shape, suggested that water flow could be characterized as matrix flow. The hydraulic properties of weathered granite at other sites followed the trends observed at our sites, implying wide applicability of the findings in this study to various types of weathered granite.

ABBREVIATIONS: KEW, Kiryu Experimental Watershed; NEW, Nishi'otafuku-Yama Experimental Watershed; WRC, water retention characteristic.

RECENT HILLSLOPE hydrology studies have suggested that bedrock, which had been conventionally regarded as impermeable (e.g., McDonnell, 1990; Soulsby, 1992; Tani, 1997), plays important roles in hydrologic and hydrochemical processes, as well as in landslide occurrence in headwater catchments. Water balance calculations conducted in small granitic catchments demonstrated that >18.2% (Terajima et al., 1993) and 35 to 55% (Kosugi et al., 2006) of annual precipitation infiltrates into bedrock. Measurements of pore water pressure (Kosugi et al., 2006) and volumetric water content (Katsura et al., 2008) and tracer experiments (Wilson et al., 1993; Noguchi et al., 1999; Frazier et al., 2002; Legout et al., 2007) also detected water flow through shallow bedrock. Water that has infiltrated into bedrock discharges very slowly into soil layers (McGlynn et al., 1999; Uchida et al., 2003; Hattanji and Onda, 2004; Kosugi et al., 2008) and controls base flow generation and chemistry (e.g., Mulholland, 1993; Burns et al., 1998; Asano et al., 2002; Soulsby

et al., 2007). Piezometric and tensiometric observations have revealed that water flows from bedrock into the overlying soil layers during storm events, implying a contribution of bedrock groundwater to landslide occurrence (Wilson and Dietrich, 1987; Terajima and Moroto, 1990; Anderson et al., 1997; Montgomery et al., 1997).

Although these previous studies have emphasized the importance of bedrock groundwater in hydrologic processes in headwater catchments, few have analyzed the water flow processes within bedrock that give rise to these phenomena. Water balance calculations and detailed observations of subsurface water interaction between soil and shallow bedrock can indeed offer credible explanations for various bedrock phenomena. A description of water flow within bedrock is indispensable, however, for evaluating the role of bedrock groundwater in runoff generation, for determining stream water chemistry, and for accurately predicting the timing, location, and scale of landslides. This, in turn, requires information on the hydraulic properties of bedrock.

Societal demands, such as oil and natural gas recovery, radioactive waste disposal, and geologic CO $_2$ storage, have encouraged the development of various methods for measuring the flow properties of tight, fresh rock samples to simulate and predict flow in deep rock layers (e.g., Brace et al., 1968; Trimmer, 1982; Finsterle and Persoff, 1997). While these methods can be successfully used for tight, fresh rock samples with extremely low porosity and permeability, their applicability to fragile, weathered bedrock is somewhat questionable. Weathered bedrock is located directly beneath soil layers and therefore has the potential to contribute greatly to water flow processes in headwater catchments. Accepted

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soil physics methods may be more reasonable for measuring its hydraulic properties.

The saturated hydraulic conductivity K_s of bedrock has been measured in situ by water injection tests (Megahan and Clayton, 1986; Kosugi et al., 2008), head recovery tests (Terajima et al., 1993; Kosugi et al., 2008), compact constant-head permeameter (Graham et al., 1997), falling-head conductivity tests in piezometers (Montgomery et al., 1997, 2002), and water application experiments (Frazier et al., 2002; Tromp-van Meerveld et al., 2007). Such in situ K_s values include the combined effects of the matrix and fractures on the saturated flow, making it impossible to evaluate the contribution of each component. Widely used models for simulating water flow through fractured bedrock, such as the dual-porosity and dual-permeability models (see the review by Šimůnek et al., 2003), require the matrix and fracture properties to be evaluated separately. For this purpose, it is desirable that the bedrock matrix K_s values are measured in a laboratory using intact bedrock core samples. Then comparison with in situ values allows evaluation of the effect of fractures on the total flow. Sampling difficulties, however, have limited the number of earlier studies that measured the core-scale (i.e., matrix) laboratory K_s values of weathered bedrock (Vepraskas et al., 1991; Johnson-Maynard et al., 1994; Hubbert et al., 2001; Katsura et al., 2006). The samples used in these studies were typically collected from shallow, highly weathered bedrock layers and did not provide sufficient information to analyze water flow through a matrix of deep, weakly weathered bedrock layers. Few studies have reported the vertical distribution of K_s including deep bedrock layers, measured using core samples from boreholes (Motyka et al., 1998; Flint, 2003; Pulido-Bosch et al., 2004). Because K_s is a fundamental hydraulic parameter, it is essential to obtain better knowledge of matrix K_s and its relation to weathering degree.

The water retention characteristic (WRC) of the bedrock matrix are also important when water flow within bedrock occurs under unsaturated conditions. Because studies have shown that weathered bedrock can serve as a medium for plants to extend their roots and extract water (e.g., Hellmers et al., 1955; Arkley, 1981; Zwieniecki and Newton, 1994; Sternberg et al., 1996; Witty et al., 2003), studies investigating the WRC of weathered bedrock have been conducted for the purpose of evaluating plant-available water capacity (e.g., Jones and Graham, 1993; Anderson et al., 1995; Hubbert et al., 2001). These studies, however, focused on plant-available water capacity rather than the water flow through weathered bedrock. They often omitted both detailed data on the wet range and the WRC of deep, weakly weathered bedrock layers that plant roots never reach. The database developed by Flint (2003) included the WRC of various volcanic rock samples from boreholes, but it lacked detailed data for the wet range. Measurements of the WRC of intact, variously weathered bedrock core samples, describing the wet range in particular, and relating the WRC to the weathering degree are important to elucidate storm runoff processes through weathered bedrock layers.

The objectives of this study were (i) to measure the hydraulic properties (i.e., K_s and WRC) of intact, weathered bedrock core samples, (ii) to examine the relationships between the hydraulic properties and the weathering degree, and (iii) to discuss the effects of fractures on total water flow. The core samples were collected from five boreholes (with depths 12–38 m) in two granitic headwater catchments in central Japan. The bedrock core

samples used for the tests were chosen at depths where various degrees of weathering were evident. The hydraulic properties of the samples were determined in the laboratory using accepted soil physics methods. In addition, we made in situ K_s measurements. By comparing these measurements with the core-scale values, we were also able to investigate the effects of fractures.

Materials and Methods

Study Sites

The bedrock core samples used in this study were collected in the Kiryu Experimental Watershed (KEW) and the Nishi'otafuku-Yama Experimental Watershed (NEW). Both watersheds are underlain by granitic bedrock.

Kiryu Experimental Watershed

The Kiryu Experimental Watershed (5.99 ha) is located in the Tanakami Mountains in Shiga Prefecture, central Japan (34°58' N, 136°00' E; Fig. 1). The elevation of the watershed ranges from 190 to 260 m above sea level. The mean annual air temperature (1997–2005) was 13.5°C, with the highest average monthly temperature (24.8°C) in August and the lowest (2.9°C) in January. The mean annual precipitation from 1972 to 2005 was 1631.0 mm, most of which fell as rain. The bedrock underlying the watershed is late Cretaceous granite, the so-called Tanakami granite (Hashimoto et al., 2000). Tanakami Granite sporadically contains pegmatite and is known to yield minerals such as topaz and crystal, but generally consists of leucocratic, equigranular, biotite granite with coarse- to medium-grained textures. The primary minerals of Tanakami granite are mainly quartz, alkali feldspars, plagioclase feldspars, and biotite (Torii, 1996).

Kosugi et al. (2006) examined the water balance in the Akakabe Watershed (WA; 0.086 ha) in KEW and an unchanneled watershed (WS; 0.024 ha) within the WA (Fig. 1). They demonstrated that 35 to 55% of annual precipitation infiltrated into the bedrock in the WS. In the WA, the bedrock infiltration accounted for only 3.5 to 7.4% of the annual precipitation. Instead, 65 to 71% of the total runoff from the WA consisted of water coming from the bedrock. Moreover, Suzuki (1980) reported that water loss by bedrock infiltration was negligible for the whole KEW. These results indicate that substantial amounts of water infiltrated into the bedrock in the WS and exfiltrated from the bedrock in the WA, and little water left KEW by flowing through deeper bedrock.

To investigate the structures and groundwater dynamics within the bedrock, boreholes were drilled at points K1, K2, and K3 (Fig. 1) using a hydraulic-feed-type boring machine. The total depths of the boreholes were 12, 20, and 15 m, respectively, and all had diameters of 6.6 cm. Undisturbed bedrock cores of 5.0-cm diameter were obtained simultaneously by drilling from the total depth of Borehole K1, the bottom 16 m of K2, and the bottom 12 m of K3. Disturbed cores were obtained from the top 4 m of Borehole K2 and the top 3 m of K3, because at these depths the application of water, which was required for collecting undisturbed cores, was stopped to minimize groundwater contamination. The disturbed cores could not be used for hydraulic property tests, but were used to examine the degree of weathering.

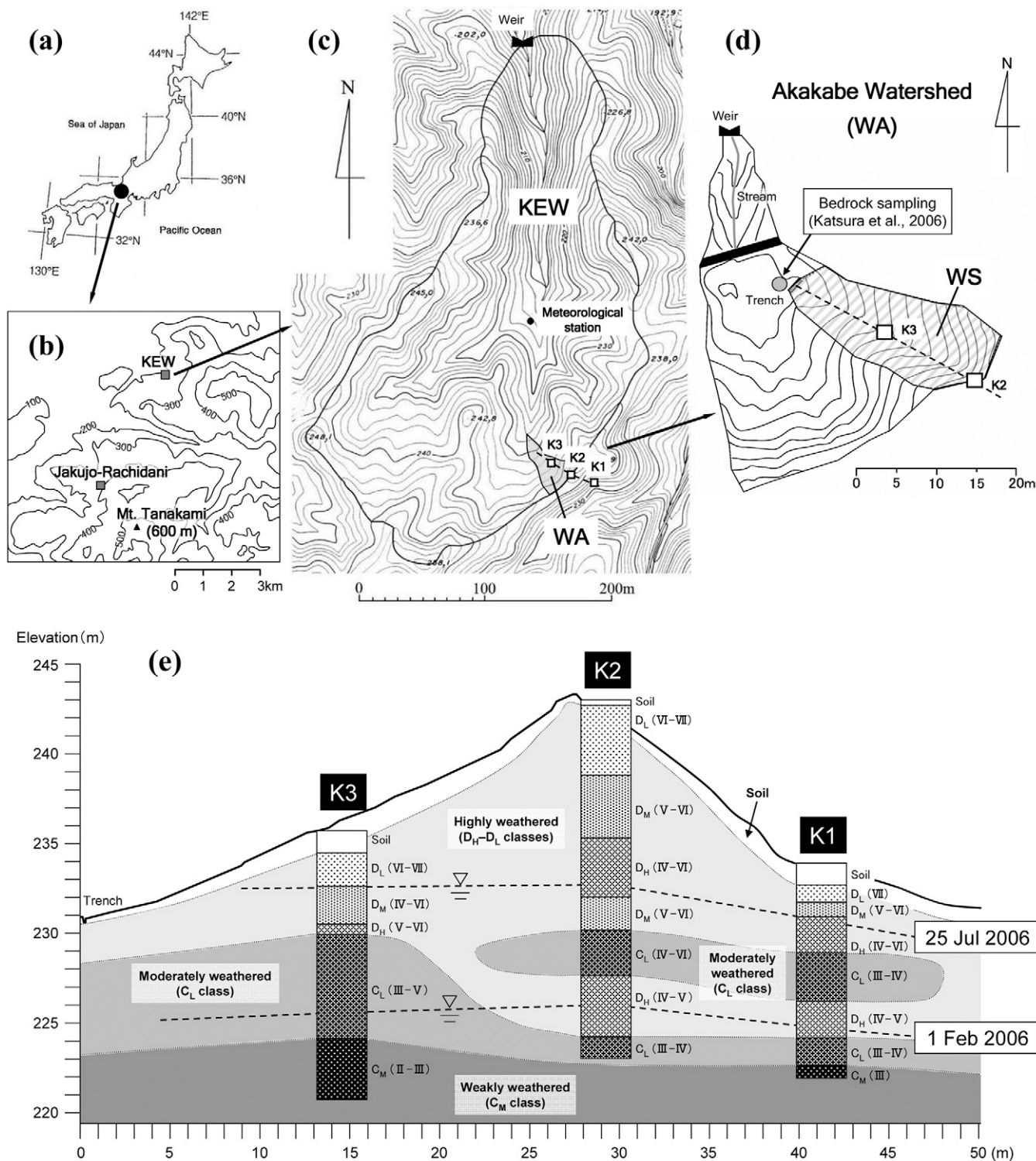


FIG. 1. Maps of (a) Japan, (b) the Tanakami Mountains, (c) the Kiryu Experimental Watershed (KEW), (d) the Akakabe Watershed (WA) with contour interval of 1 m, and (e) the weathering class and core shape (Fig. 3) profile observed in boreholes K1 to K3 and estimated for the range indicated by a dashed line in (c) and (d), including the highest and lowest bedrock groundwater tables observed in 2006.

Nishi'otafuku-Yama Experimental Watershed

The Nishi'otafuku-Yama Experimental Watershed (1.87 ha) is a headwater catchment in the Rokko Mountains (34°46' N, 135°16' E) in Hyogo Prefecture, central Japan (Fig. 2). The watershed is located at the edge of an elevated peneplain extending around the summit of Mt. Rokko (932 m) and ranges from 740 to 875 m above sea level, with a mean gradient of 35° and mean

annual precipitation of approximately 1800 mm. The bedrock is equigranular, Cretaceous granite, characterized by light pink feldspar, with late Cretaceous to Paleogene quartz porphyry dykes intruding in places. This granite is called Rokko granite and contains the primary minerals quartz, plagioclase feldspars, alkali feldspars, and biotite (Kasama, 1968).

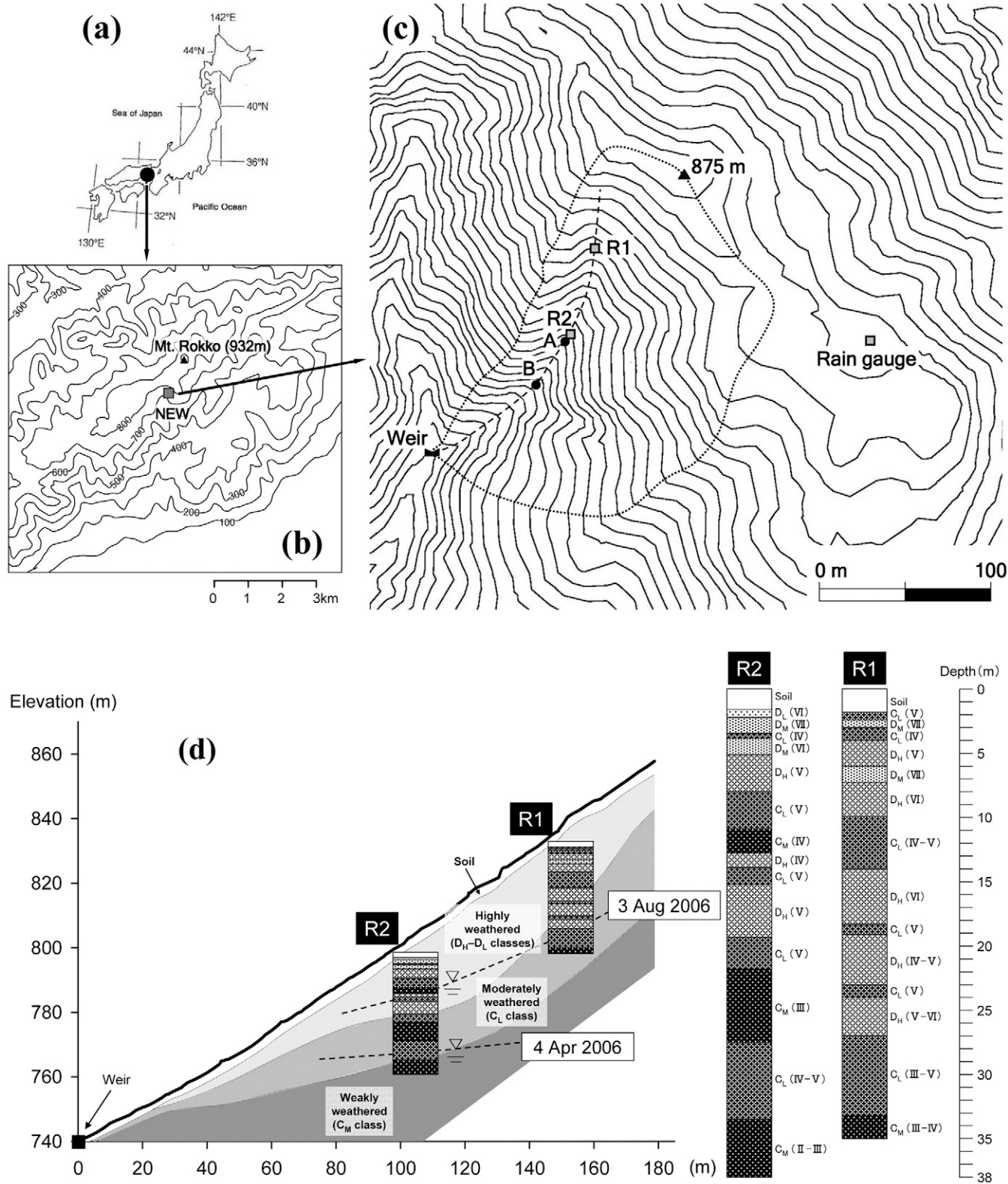


FIG. 2. Maps of (a) Japan, (b) the Rokko Mountains, (c) the Nishi'otafuku-Yama Experimental Watershed (NEW) with a contour interval of 5 m and watershed boundary represented by the dotted line, and (d) the weathering class and core shape (Fig. 3) profiles observed in boreholes R1 and R2 and estimated for the range indicated by a dashed line in (c), including the highest and lowest bedrock groundwater tables observed in 2006.

Field observations performed in NEW have revealed that the groundwater in soil layers at Point B in Fig. 2c occasionally remained formed for several months (Kosugi et al., 2008). This behavior could not be explained by the physical model based on

soil water movement. Using geochemical and geothermal evidence, Kosugi et al. (2008) demonstrated that the source of this type of groundwater is the deep bedrock groundwater.

To investigate how deep bedrock groundwater affects the groundwater in soil layers, boreholes were drilled at Points R1 and R2 (Fig. 2c) using a hydraulic-feed-type boring machine. The total depths of the boreholes were 35 and 38 m, respectively, and both had diameters of 6.6 cm. Undisturbed bedrock cores of 5.0-cm diameter were obtained simultaneously by drilling from the total depths of both boreholes.

Weathering Class and Core Shape

Granite cores are generally classified into four categories—A, B, C, and D—according to the weathering degree (Japanese Society of Soil Mechanics and Foundation Engineering, 1985). Table 1 summarizes the characteristics of each weathering class. The weathering class A represents fresh, unweathered granite, and the weathering class increases with weathering degree. Both weathering classes C and D are divided into three subcategories, with subscripts H, M, and L.

Figures 1e and 2d show the weathering class profiles observed for the total depths of the boreholes in KEW and NEW, respectively. The bedrock core samples from these boreholes ranged in weathering class from C_M to D_L . Although less-weathered bedrock cores were observed at some depths, the shallower bedrock layer was generally more highly weathered. Based on the weathering class profiles observed in the boreholes and the results of the resistivity image profiling, the weathering degree distributions were roughly estimated in Fig. 1e and 2d for the ranges indicated by the dashed lines in Fig. 1c and 2c, respectively. Here, highly, moderately, and weakly weathered bedrock is defined as classes D_H to D_L , C_L , and C_M , respectively.

Figures 1e and 2d also show the core shape of each zone divided by the weathering class. The core shape is represented by the Roman numerals I to VII according to the core length, or fracture density, and the core material (Japan Construction Information Center, 1986; Fig. 3). The figures show that, for both sites, the core shape was generally VI to VII for the D_L class, V to VII for the D_M class, IV to VI for the D_H class, IV to V for the C_L class, and II to IV for the C_M class.

Figures 1e and 2d include the highest and lowest bedrock groundwater tables observed in 2006, estimated using the groundwater level records in the boreholes. The highest and lowest groundwater tables were observed on 25 July and 1 February in KEW, and 3 August and 4 April in NEW, respectively. The figures demonstrate that thick, unsaturated bedrock layers existed just

beneath the soil layers. The unsaturated bedrock layers attained a thickness of >10 m at Point K2 in KEW (Fig. 1e), and >28 m at Point R1 in NEW (Fig. 2d). These groundwater level measurements emphasize the importance of saturated and unsaturated hydraulic properties in water flow processes through the weathered bedrock.

Preparation of Core Samples for the Hydraulic Properties Tests

Core samples for the hydraulic properties tests were prepared following the methods described by Katsura et al. (2006). In the laboratory, one to three samples, more than approximately 5 cm in length without apparent fractures and sheeting joints, were chosen from each zone characterized by the weathering classes, as shown in Fig. 1e and 2d. The top and bottom of the samples were carefully trimmed so that the sample became a column. No samples could be prepared from some zones because (i) the lack of applied water disturbed the cores during borehole drilling (shallow depths in Boreholes K2 and K3), (ii) apparently intact samples of more than approximately 5 cm were obtained but were split or broken, typically along invisible minor fractures, or (iii) cores had been disturbed due to the poor condition of the drilling machine. Consequently, 5, 6, 5, 13, and 10 samples were prepared from Boreholes K1, K2, K3, R1, and R2, respectively. The trimmed samples were individually placed in the center of acrylic cylinders of inner diameter 5.8 cm. Liquid silicone rubber (TSE350, GE Toshiba Silicones, Tokyo) was poured into the space between the bedrock sample and the cylinder wall to the top of the sample. After 1 d, when solidification of the liquid silicone rubber was complete, any silicone rubber adhering to the bottom of the samples was carefully removed. Because infiltration of the silicone rubber was not observed and solidified silicone rubber does not allow water to pass through it, accurate measurement of the hydraulic properties was thus possible (Katsura et al., 2006).

Saturated Hydraulic Conductivity Tests

All core samples were gradually saturated with water during 72 h from the bottom upward to avoid air entrapment. For some samples that were not readily saturated, we also used a vacuum pump to ensure full saturation. The falling-head test (Reynolds et al., 2002) was performed on all samples. The times t_i for the water level to fall from the initial water level h_0 to the predetermined

TABLE 1. Classification of weathering class (Japanese Society of Soil Mechanics and Foundation Engineering, 1985).

Weathering class	Degree of weathering	Color	Hardness
A	fresh	bluish gray to milk-whitish gray	very hard, emits a clank when hammered
B	generally fresh	milk-whitish gray to (light) brownish gray	hard, emits a light clank when hammered
C_H	weathered along fractures, plagioclase and biotite are partly altered	brownish gray to (light) grayish brown	moderately hard, emits a dull clank when hammered
C_M	weathered except inside rock body, plagioclase and biotite are altered further	grayish brown to light yellowish brown	slightly soft to hard, easily split when hammered
C_L	weathered entirely while retaining rock fabric and structure, plagioclase is generally altered, quartz is left unweathered	light yellowish brown to yellowish brown	soft, partly crushed by finger pressure
D_H	entirely weathered, plagioclase is altered into clay	yellowish brown	soft, crushed by finger pressure
D_M	entirely weathered, most of feldspar is generally altered	yellowish brown	very soft, crushed by finger pressure
D_L	entirely weathered, most of feldspar is generally altered into clay, quartz is decomposed into fine grains	yellowish brown	very soft, powdered by finger pressure

	Schematic diagram	Core shape
I		Stick-shaped core more than 50 cm in length
II		Stick-shaped core 15–50 cm in length
III		Stick- to fragment-shaped core 15–5 cm in length
IV		Fragment-shaped core less than 5 cm in length
V		Core mainly consisting of breccia
VI		Core mainly consisting of sand
VII		Core mainly consisting of clay

FIG. 3. Classification of the core shape (Japan Construction Information Center, 1986).

levels h_i were recorded ($i = 1$ to n , $n = 3-5$). The K_s was determined by fitting the following equation to the observed h_i-t_i data (Reynolds et al., 2002):

$$h(t) = \exp\left(\ln h_0 - K_s \frac{St}{aL}\right) \quad [1]$$

where t is the time, h is the water level, a is the cross-sectional area of the falling-head standpipe (0.5 cm^2), and S and L are the cross-sectional area (19.6 cm^2) and height of the sample, respectively. This method of determining K_s has been shown to have remarkably small error in the determined K_s value (Katsura et al., 2006).

For comparison with these core-scale K_s values, we also made in situ K_s measurements at several depths in the boreholes using head recovery and water injection techniques (Japanese Geotechnical Society, 2004).

Water Retention Tests

The water retention tests were conducted by the pressure plate method (Dane and Hopmans, 2002). After being fully saturated, the core samples were placed on a previously saturated, porous ceramic plate. Because some samples had space between the bottom of the bedrock and the porous plate, saturated quartz sand (0.17–0.32 mm in diameter) was used to fill the space to ensure continuous water flow from the bedrock sample to the porous plate. The WRCs were measured by setting the air pressure at 10, 20, 30, 50, 70, 100, 200, 500, and 1000 cm. For the samples for which quartz sand was used to fill the space, the measured data were corrected to obtain the WRC of the bedrock sample by using the independently measured WRC of pure quartz sand (Katsura et al., 2006). To allow additional tests for measuring other properties, such as the unsaturated hydraulic conductivity and solute dispersion coefficient, on these bedrock samples, the saturated volumetric water content of a sample taken at nearby depths in the same weathering class zone as that of the sample used in the hydraulic properties tests was used to determine the WRC of each sample.

To characterize the observed WRCs, we used the lognormal model (Kosugi, 1996), which was based on an inverse relationship (capillary rise function) between the pressure head, ψ , and the pore radius, r :

$$\psi = A/r \quad [2]$$

where A is a constant with an approximate value of -0.149 cm^2 . By assuming the pore radius to be lognormally distributed (Kosugi, 1996),

$$g(r) = \frac{\theta_s - \theta_r}{\sqrt{2\pi}\sigma r} \exp\left\{-\frac{[\ln(r/r_m)]^2}{2\sigma^2}\right\} \quad [3]$$

we obtained

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} = Q\left[\frac{\ln(\psi/\psi_m)}{\sigma}\right] \quad [4]$$

where S_e is the effective saturation, θ is the volumetric water content, and θ_s and θ_r are the saturated and residual volumetric water contents, respectively. The parameter r_m is the medium pore radius, ψ_m is the pressure head at $S_e = 0.5$, which is related to r_m by Eq. [2], σ is a dimensionless parameter characterizing the width of the pore-size distribution, and Q is the complementary normal distribution function:

$$Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du \quad [5]$$

The lognormal model expressed by Eq. [4] adequately describes the observed WRC of many kinds of soils (Kosugi, 1996) and we assumed that it is applicable to weathered bedrock.

While other model equations for WRC often use empirical parameters (e.g., van Genuchten, 1980), all of the parameters in Eq. [4] are physically based. This allows physically significant comparison of the WRCs among the samples. Determination of the parameter values of θ_s , θ_r , ψ_m , and σ requires Eq. [4] to be fitted to the observed WRC. This procedure involves minimizing the residual sum of squares, computed using the measured and calculated values of θ at the same ψ . We initially fitted all four parameters in Eq. [4], but found nonunique solutions for some samples. In these cases, θ_s was fixed at the measured value while θ_r was set at the lowest observed θ and only ψ_m and σ were optimized.

Results

Saturated Hydraulic Conductivity

Figures 4 and 5 show the vertical distribution of the core-scale K_s in each borehole in KEW and NEW, respectively. For illustration in Fig. 4 and 5, it was assumed that the K_s of the core sample used for the test represented the whole zone divided by the weathering classes from which the sample was taken. While Borehole K3 showed a consistent decrease in core-scale K_s with increasing depth, the core-scale K_s changed little for depths between 1.23 and 7.70 m in Borehole K1 and for depths >4.20 m in K2 (Fig. 4). In addition, the core-scale K_s showed no systematic change with depth in Boreholes R1 and R2 (Fig. 5). Instead, the core-scale K_s dropped sharply at the C_M -class bedrock. The apparently consistent decrease in K3 can also be attributed to the greater degree of weathering in the shallower layer (C_L class) than in the deeper layer (C_M class). Thus, the weathering class, rather than the depth, is more relevant to the core-scale K_s in both KEW and NEW.

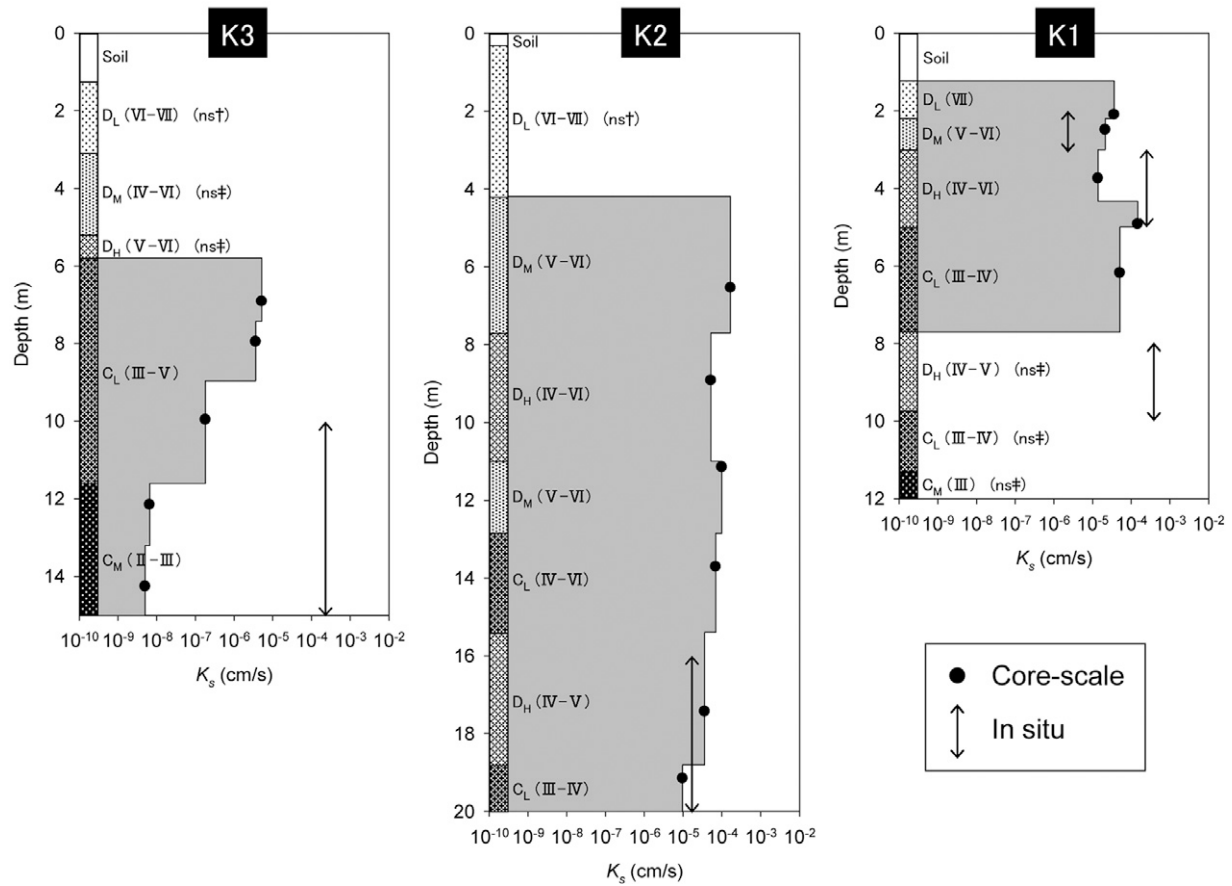


FIG. 4. Vertical distribution of saturated hydraulic conductivity (K_s) and the core shape observed in the boreholes in the Kiryu Experimental Watershed; ns†, no samples could be prepared because no water application during the borehole drilling disturbed the cores; ns‡, no samples could be prepared because apparently intact cores were obtained but were split or broken, typically along invisible minor fractures often clogged with clayey materials, for the total depth of the weathering class zone while being trimmed.

The results of the in situ K_s measurements are also shown in Fig. 4 and 5, where arrows indicate the measurement depths. The measurement sections ranged in weathering class from D_L to C_M , but the range of the obtained values (1.2×10^{-6} – 3.8×10^{-4} cm s⁻¹) was narrow compared with that of the core-scale measurements. Since in situ values include the effects of both the matrix and the fractures, we expected that the in situ measurements would be larger than the core-scale values. Indeed, the in situ values for the C_M -class bedrock were much larger than the core-scale values. For the D_L - to C_L -class bedrock, however, we observed no such trend; the in situ values were generally similar to the core-scale values.

Water Retention Characteristics

Figure 6 displays the WRCs of samples for each weathering class. Note that Fig. 6b, which shows the results for NEW, includes no curve for the D_L -class sample because no D_L -class sample could be prepared from this site. The WRCs of the samples with weathering classes D_L to C_L showed a gradual decrease in θ in the wet range where ψ was greater than -200 cm (corresponding to $r = 7.45 \times 10^{-4}$ cm). In the dry range, where ψ was less than -200 cm, θ decreased little with decreasing ψ . Although the absolute value of θ at the same ψ value was different among the samples, the curves were nearly parallel to each other. The C_M -class samples, on the other hand, showed almost no change in θ across the entire range of applied ψ . These trends

were common to both KEW and NEW, and were observed for all the other samples.

Figure 6 includes the curves of the model Eq. [4] fitted to the observed WRC. We did not apply Eq. [4] to the C_M -class samples because they exhibited no substantial change in θ . Figure 6 highlights the close agreement between the observed and fitted curves of the D_L - to C_L -class samples, suggesting that Eq. [4] adequately described the observed WRC of these samples. We also found close matches between the observed and fitted curves for all the D_L - to C_L -class samples using individually optimized parameters.

Figures 7 and 8 show the vertical distribution of θ_s , θ_r , and the effective porosity, θ_e , in each borehole in KEW and NEW, respectively. Here, θ_e is defined as the difference between θ_s and θ_r , and indicates the amount of porosity that effectively contributes to water flow. For the C_M -class samples, to which Eq. [4] was not applied, the measured value was substituted for θ_s , and θ observed at the lowest ψ (approximately -1000 cm) was used as θ_r . In Fig. 7 and 8, it was again assumed that the water retention parameters of the sample used for the test represented the whole zone defined by the weathering classes from which the sample was taken.

Although all of the parameter values showed no systematic change with increasing depth in Boreholes K1 and K2, a consistent decrease in θ_s and θ_r with depth was found in Borehole K3 (Fig. 7). The value of θ_e in K2 tended to be larger than that in K1 and K3 to greater depths. In Borehole R1, both θ_s and θ_r

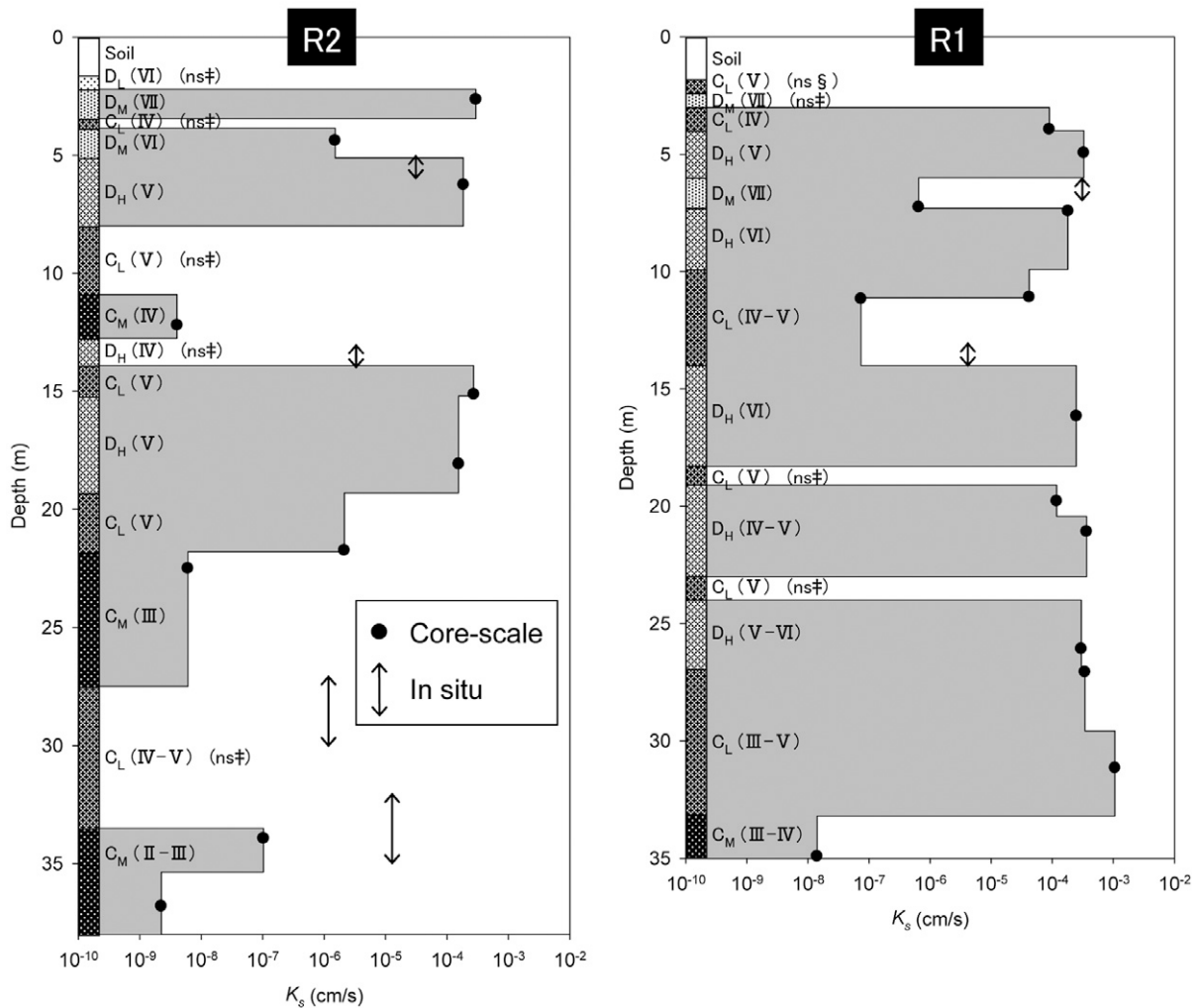


FIG. 5. Vertical distribution of saturated hydraulic conductivity (K_s) and the core shape observed in the boreholes in the Nishi'otafuku-Yama Experimental Watershed; ns#, no samples could be prepared because apparently intact cores were obtained but were split or broken, typically along invisible minor fractures often clogged with clayey materials, for the total depth of the weathering class zone while being trimmed. ns§, no samples could be prepared because cores were disturbed due to the poor condition of the drilling machine.

showed less change with depth than in Borehole R2 (Fig. 8). The value of θ_c was large to greater depths in R1 than in R2. While no consistent changes in θ_s , θ_p , and θ_c were observed in K1, K2, and R2, θ_c was extremely small in the C_M -class bedrock. In addition, larger values of θ_c at greater depths were observed in K2 and R1, probably because the C_M -class bedrock was absent and occurred only at the greatest depth, respectively. The consistent decrease in these parameters in K3 can also be attributed to the greater degree of weathering of shallower layers. Thus, as observed for K_s , the weathering class, rather than the depth, was more relevant to θ_c in both KEW and NEW.

Discussion

Changes in Core-Scale Hydraulic Properties with Weathering Progress

As shown above, weathering class rather than depth appeared to determine the core-scale hydraulic properties of the bedrock in both sites. Unfortunately, of all 39 samples prepared in this study, only one sample could be used as a D_L -class sample. To supplement the insufficient D_L -class data, hydraulic properties

of a D_L -class sample (approximately 1860 cm³ in volume), taken at the point indicated by the gray circle in Fig. 1d (Katsura et al., 2006), were also used in the following figures and discussion. Izumiyama (2007) also measured the hydraulic properties of five D_L -class bedrock samples (approximately 100 cm³ in volume) manually collected from 0- to 30-cm depths from the bedrock surface in the Jakujo-Rachidani Experimental Watershed in the Tanakami Mountains, located about 5 km southwest of KEW (Fig. 1b). These samples were Tanakami granite, and their hydraulic properties were used as the D_L -class properties. Furthermore, hydraulic properties of undisturbed soil samples (100 cm³ in volume), taken from depths within 40 cm from the ground surface in KEW ($n = 4$) and within 50 cm in NEW ($n = 7$), were also used for comparison with the bedrock hydraulic properties.

Saturated Hydraulic Conductivity

Figure 9 shows the relationship between weathering class and K_s . The attached table shows the significance level of the difference between the mean values of the corresponding categories calculated using a two-tailed t -test. Table 2 summarizes the geometric mean K_s for each weathering class. While the soils had K_s

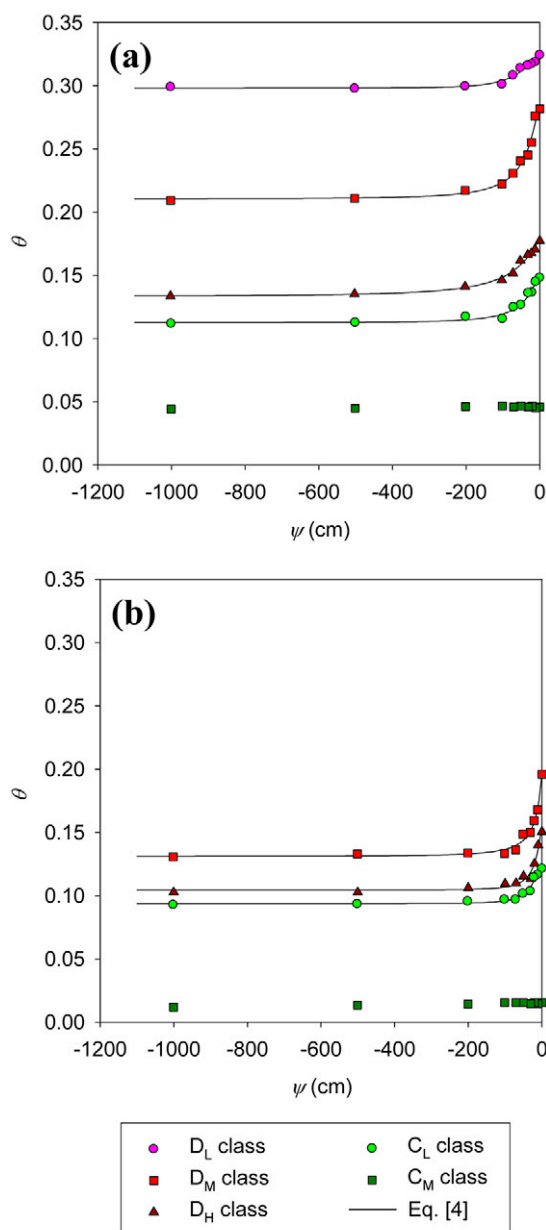


FIG. 6. Water retention curves of representative samples from (a) the Kiryu Experimental Watershed and (b) the Nishi'otafuku-Yama Experimental Watershed. C_M through D_L indicate weathering classes: C_M , easily split when hammered, plagioclase and biotite altered; C_L , partly crushed by finger pressure, plagioclase altered and quartz left unweathered; D_H , crushed by finger pressure, plagioclase altered into clay; D_M , crushed by finger pressure, most of feldspar generally altered; D_L , powdered by finger pressure, most of feldspar generally altered into clay and quartz decomposed into fine grains.

values $>1.1 \times 10^{-2} \text{ cm s}^{-1}$, all of the bedrock samples had values $<1.7 \times 10^{-3} \text{ cm s}^{-1}$. Within the same weathering class, although the values of the soil and D_H -class bedrock of Tanakami granite were significantly lower ($P < 0.05$) and higher ($P < 0.01$) than those of Rokko granite, respectively, the difference in the geometric mean K_s between Tanakami granite and Rokko granite was not so large for all weathering classes. Based on Fig. 9 and Table 2, three groups could be defined for both Tanakami granite and Rokko granite: (i) C_M -class bedrock, which had core-scale K_s significantly lower than any other weathering class ($P < 0.01$)

with a geometric mean of $8.3 \times 10^{-9} \text{ cm s}^{-1}$, (ii) C_L - to D_L -class bedrock, which showed less significant difference in core-scale K_s among these weathering classes with a geometric mean ranging from 1.7×10^{-5} to $3.4 \times 10^{-4} \text{ cm s}^{-1}$, and (iii) soil, which had core-scale K_s significantly higher than any other weathering class ($P < 0.001$) with a geometric mean of $6.9 \times 10^{-2} \text{ cm s}^{-1}$.

Water Retention Characteristics

Figure 10 shows the relationship between weathering class and θ_s , θ_r , and θ_e , the arithmetic means of which are summarized in Table 2. Figure 10a demonstrates that θ_s increased with the weathering degree, including for soil and both Tanakami granite and Rokko granite. Because the θ_r of the bedrock also tended to increase at a similar rate to θ_s with increasing weathering degree (Fig. 10b), however, the θ_e of the bedrock did not change greatly with weathering class (Fig. 10c). In fact, the differences in θ_e among weathering classes of C_L through D_L were generally not significant. This suggests that the consistent increase in θ_s of the bedrock was derived from an almost equal increase in θ_r and therefore contributed little to the increase in θ_e from the weathering classes of C_L into D_L . The C_M -class bedrock, however, generally had significantly smaller θ_e than any other weathering class for both Tanakami granite and Rokko granite. In addition, the soil had significantly larger θ_e than the bedrock of any weathering class ($P < 0.01$). Except for the soil that had θ_s and θ_r significantly smaller for Tanakami granite than for Rokko granite ($P < 0.01$ and $P < 0.001$, respectively), no significant differences in θ_s , θ_r , and θ_e were observed between the Tanakami granite and Rokko granite.

Figure 11 shows the relationships between weathering class and ψ_m and σ , with the mean values summarized in Table 2. Whereas the Tanakami granite exhibited significantly larger ψ_m (i.e., larger r_m ; Eq. [2]) and σ of the soil than of the bedrock ($P < 0.01$), no significant differences in either ψ_m or σ among weathering classes were observed for the Rokko granite. In addition, there were no significant differences between the Tanakami granite and Rokko granite except for ψ_m of the D_H -class bedrock ($P < 0.01$) and σ of the soil ($P < 0.05$).

Using the mean values of the retention parameters (Table 2), water retention curves for each weathering class are shown in Fig. 12. The curve of the C_M -class bedrock, for which the mean ψ_m and σ could not be determined, was drawn by arbitrarily using values of $\psi_m = -30.0 \text{ cm}$ and $\sigma = 1.30$ (we confirmed that selection of the ψ_m and σ values did not strongly affect the shape of the curve). For each weathering class, although the absolute value of θ at the same ψ differed between Tanakami granite and Rokko granite, the mean water retention curves were nearly parallel to each other. This trend also held for the soil, despite the significantly smaller θ_s and θ_r for Tanakami granite than for Rokko granite, as was reflected in the lack of significant difference in θ_e and ψ_m . This difference, which should have been derived from the significant difference in σ ($P < 0.05$) between Tanakami granite and Rokko granite (Fig. 10b), was hidden by the parallelism that the mean curves exhibited. Likewise, the significant difference in ψ_m of the D_H -class bedrock between Tanakami granite and Rokko granite ($P < 0.01$) did not greatly affect the parallelism of the mean curves. Because it is not the WRC but the water-capacity function, defined as $d\theta/d\psi$, that controls unsaturated water flow, we conclude that Tanakami granite and Rokko

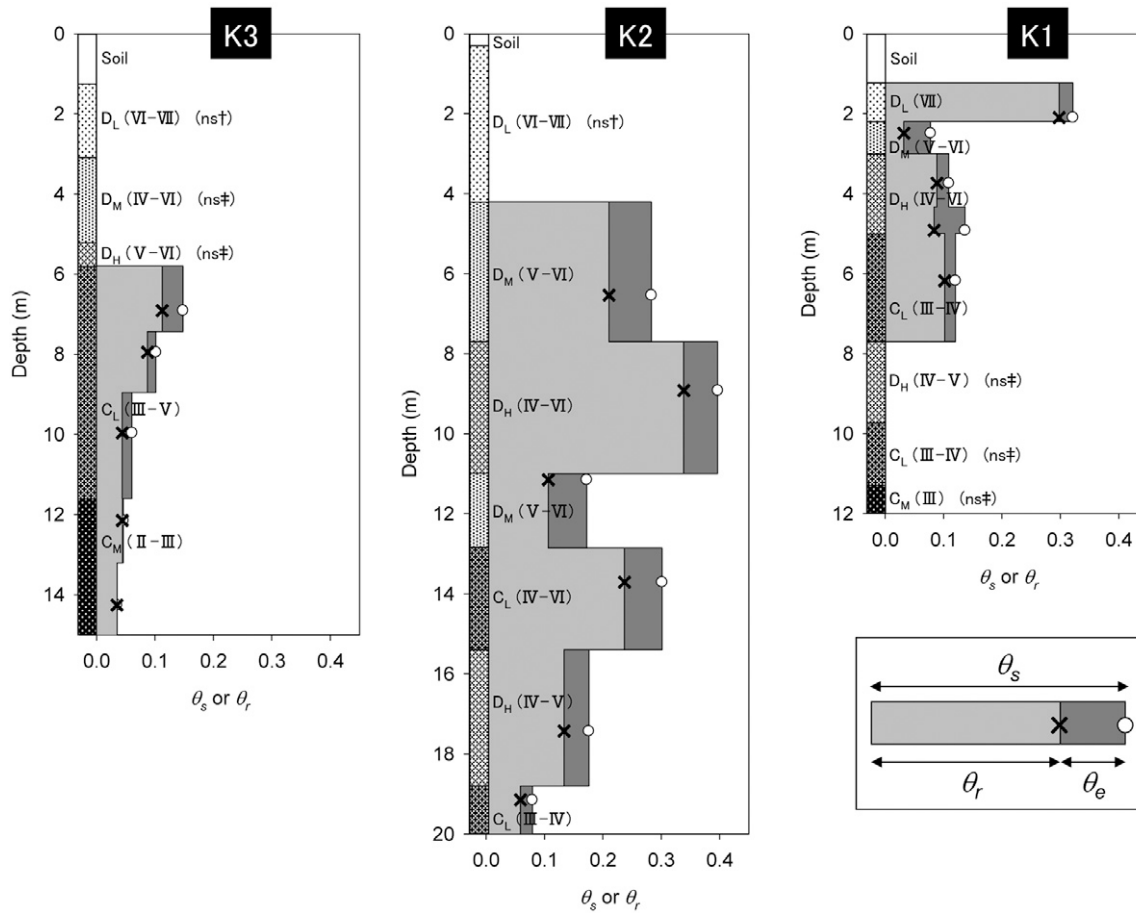


FIG. 7. Vertical distribution of saturated and residual volumetric water content (θ_s and θ_r , respectively), effective porosity (θ_e), and the core shape observed for the boreholes in the Kiryu Experimental Watershed; ns†, no samples could be prepared because no water application during the borehole drilling disturbed the cores; ns‡, no samples could be prepared because apparently intact cores were obtained but were split or broken, typically along invisible minor fractures often clogged with clayey materials, for the total depth of the weathering class zone while being trimmed.

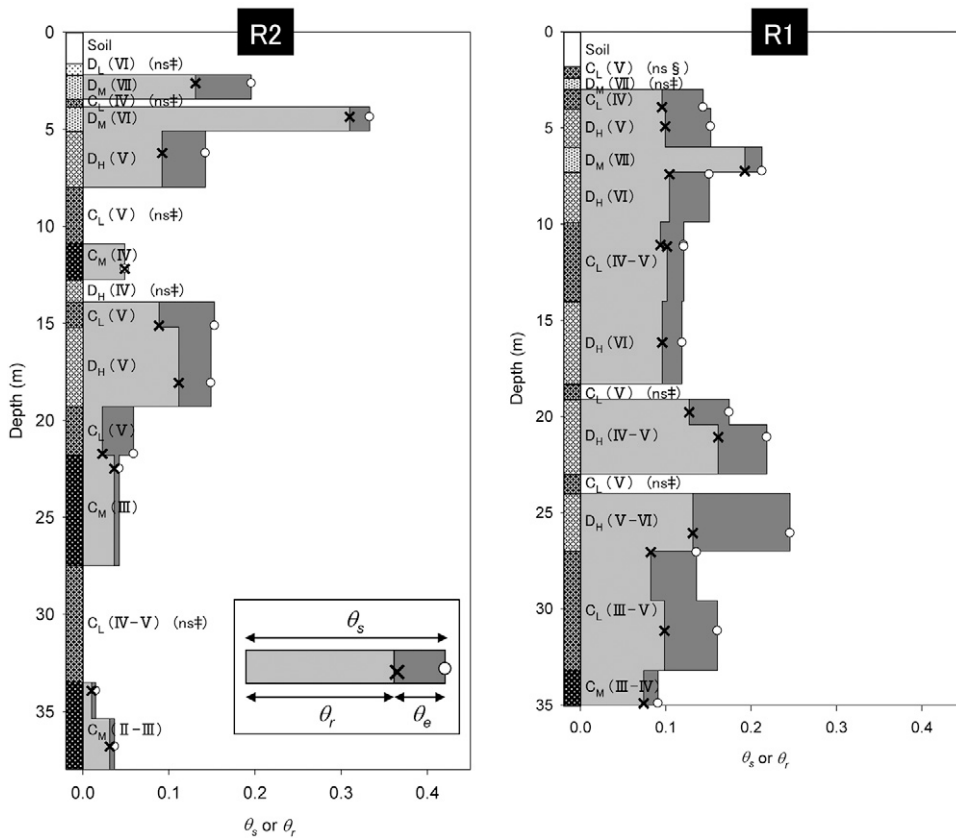


FIG. 8. Vertical distribution of saturated and residual volumetric water content (θ_s and θ_r , respectively), effective porosity (θ_e), and the core shape observed for the boreholes in the Nishi'otafuku-Yama Experimental Watershed; ns‡, no samples could be prepared because apparently intact cores were obtained but were split or broken, typically along invisible minor fractures often clogged with clayey materials, for the total depth of the weathering class zone while being trimmed; ns§, no samples could be prepared because cores were disturbed due to the poor condition of the drilling machine.

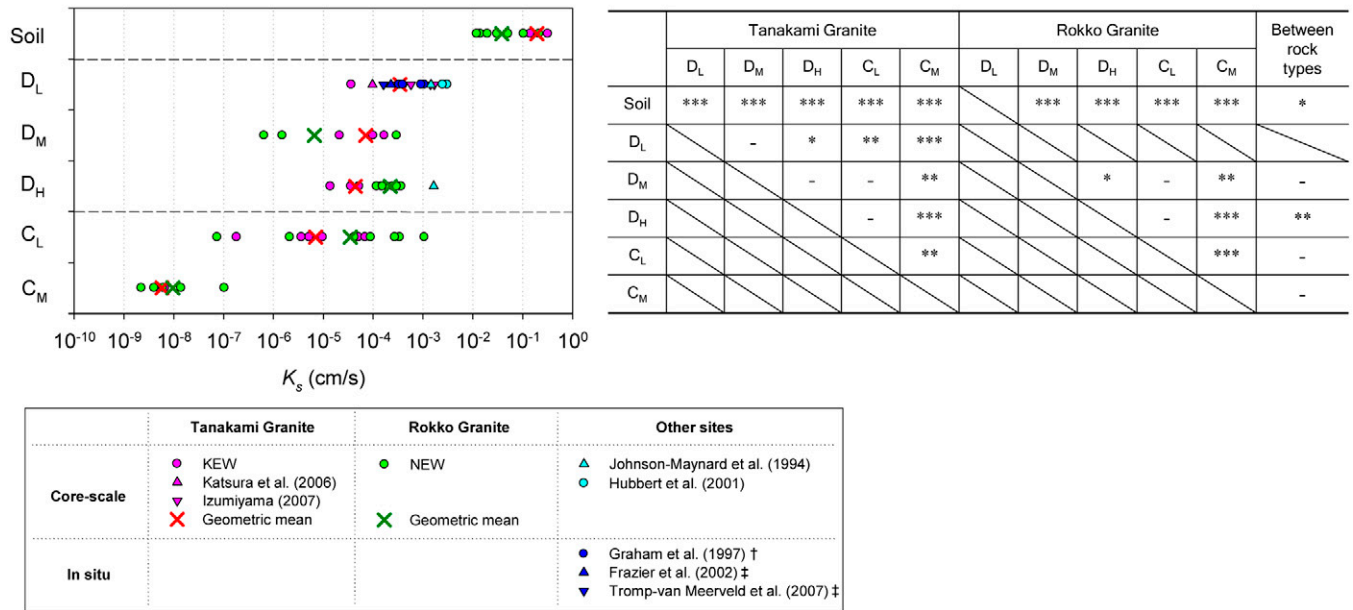


FIG. 9. Relationship between weathering class (see Table 1 for weathering class definitions) and saturated hydraulic conductivity (K_s) at the Kiryu Experimental Watershed (KEW), the Nishi'otafuku-Yama Experimental Watershed (NEW), and other sites. The table to the right shows the significance level of the difference between the mean values of the corresponding categories calculated using a two-tailed t -test: ***significant at $P < 0.001$, **significant at $P < 0.01$, *significant at $P < 0.05$, dash signifies not significant. † Estimated using a compact constant-head permeameter. ‡ Final infiltration rate is regarded as K_s .

TABLE 2. Mean parameter values for the core-scale hydraulic properties of saturated hydraulic conductivity (K_s), saturated and residual volumetric water contents (θ_s and θ_r , respectively), effective porosity (θ_e), pressure head when the effective saturation (S_e) is 0.5 (ψ_m), and the parameter characterizing the width of the pore size distribution (σ).

Weathering class†	Tanakami granite						Rokko granite						Total					
	$K_s^\ddagger$	$\theta_s^\S$	$\theta_r^\S$	$\theta_e^\S$	$\psi_m^\ddagger$	$\sigma^\S$	$K_s^\ddagger$	$\theta_s^\S$	$\theta_r^\S$	$\theta_e^\S$	$\psi_m^\ddagger$	$\sigma^\S$	$K_s^\ddagger$	$\theta_s^\S$	$\theta_r^\S$	$\theta_e^\S$	$\psi_m^\ddagger$	$\sigma^\S$
	cm s ⁻¹				cm		cm s ⁻¹				cm		cm s ⁻¹				cm	
Soil	1.9×10^{-1}	0.509	0.152	0.357	-14.0	2.41	3.8×10^{-2}	0.694	0.359	0.333	-20.6	1.51	6.9×10^{-2}	0.626	0.284	0.342	-17.9	1.84
D _L	3.4×10^{-4}	0.291	0.244	0.047	-29.4	0.74	-	-	-	-	-	-	-	-	-	-	-	-
D _M	7.1×10^{-5}	0.178	0.116	0.061	-41.5	1.09	6.6×10^{-6}	0.247	0.211	0.036	-28.4	1.90	2.2×10^{-5}	0.212	0.164	0.049	-34.3	1.49
D _H	4.4×10^{-5}	0.205	0.161	0.044	-52.4	1.15	2.2×10^{-4}	0.169	0.116	0.054	-21.8	1.15	1.3×10^{-4}	0.181	0.131	0.050	-29.2	1.15
C _L	7.0×10^{-6}	0.135	0.107	0.028	-52.9	1.34	3.4×10^{-5}	0.128	0.083	0.045	-35.5	1.49	1.7×10^{-5}	0.131	0.094	0.037	-42.7	1.42
C _M	5.8×10^{-9}	0.040	0.040	0.001	-	-	9.6×10^{-9}	0.047	0.040	0.007	-	-	8.3×10^{-9}	0.045	0.040	0.005	-	-

† See Table 1 for weathering class definitions.

‡ Geometric mean.

§ Arithmetic mean.

granite with the same weathering class can be treated equally in these cases. Moreover, as suggested by generally no significant differences in θ_e , ψ_m , and σ among the C_L- to D_L-class bedrock, the mean water retention curves of these weathering classes were also nearly parallel to each other. Therefore, the three groups, which were defined by the core-scale K_s , could also be applied to the WRC as follows: (i) for C_M-class bedrock, θ changes little across a wide range of ψ values; (ii) for C_L- to D_L-class bedrock, θ changes gradually in the wet range (ψ greater than -200 cm) and little in the dry range (ψ less than -200 cm)—the water retention curves are nearly parallel to each other; (iii) for soil, θ exhibits a large change across a wide range of ψ values, with the change in the wet range especially large.

Relationship between Core-Scale Saturated Hydraulic Conductivity and Water Retention Parameters

Figure 13 displays the relationships between the core-scale K_s and the water retention parameters (i.e., θ_s , θ_r , θ_e , ψ_m , and σ).

The values of K_s and ψ_m were plotted on a logarithmic scale, and Fig. 13d and 13e, which show the relationship of K_s with ψ_m and σ , respectively, do not include the values for the C_M-class samples. We found positive correlations between $\log K_s$ and θ_s (Fig. 13a), θ_r (Fig. 13b), θ_e (Fig. 13c), and $-\log |\psi_m|$ (Fig. 13d). Calculation of the coefficient of determination, R^2 , suggested that the correlation with $\log K_s$ was stronger for θ_s ($R^2 = 0.60$) than for θ_r ($R^2 = 0.36$), θ_e ($R^2 = 0.57$), $-\log |\psi_m|$ ($R^2 = 0.46$), and σ ($R^2 = 0$), although a nonlinear relationship between θ_e and $\log K_s$ (Fig. 13c) inevitably degraded the correlation. Instead, for the $\log \theta_e$ - $\log K_s$ plot (Fig. 13c'), a very strong correlation was found ($R^2 = 0.83$). Because θ_e contained information derived from θ_s and θ_r , we can conclude that θ_e and ψ_m are the most important parameters for determining the core-scale K_s .

Figure 14 shows the relationship between the observed $\log \theta_e$ and $-\log |\psi_m|$. The plot was classified according to the value of $\log K_s$. As suggested in Fig. 13, for both the bedrock and soil, K_s became larger as θ_e increased. Moreover, for the same θ_e , a larger

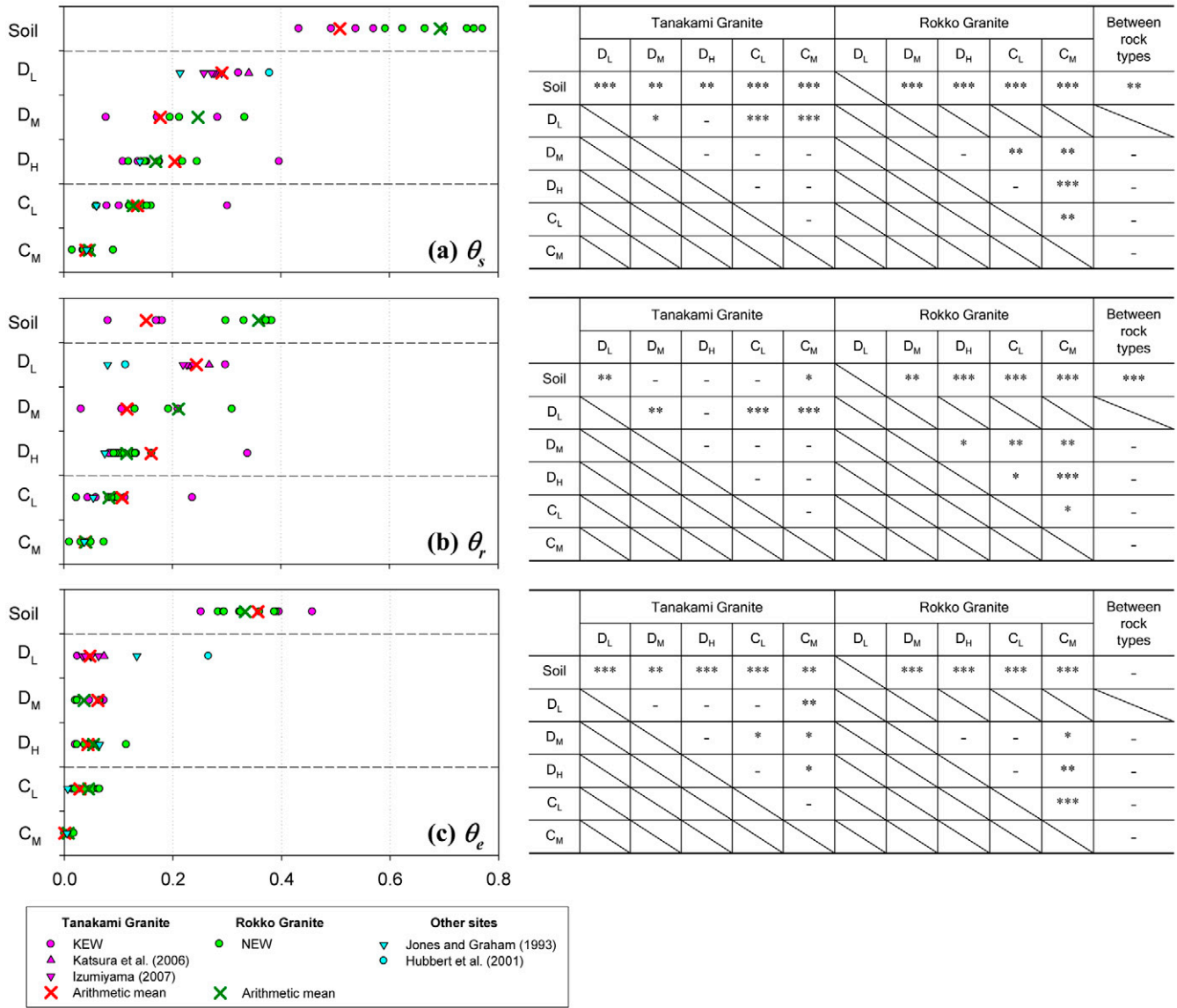


FIG. 10. Relationship between weathering class (see Table 1 for weathering class definitions) and (a) saturated volumetric water content (θ_s), (b) residual volumetric water content (θ_r), and (c) effective porosity (θ_e) at the Kiryu Experimental Watershed (KEW), the Nishi'otafuku-Yama Experimental Watershed (NEW), and other sites. The tables to the right show the significance level of the difference between the mean values of the corresponding categories: ***significant at $P < 0.001$, **significant at $P < 0.01$, *significant at $P < 0.05$, dash indicates not significant.

value of $-\log |\psi_m|$ led to larger K_s . Thus, Fig. 14 demonstrates that $\log K_s$ is closely represented by $\log \theta_e$ and $-\log |\psi_m|$.

Based on the model of Mualem (1976), Kosugi (1997) derived a relationship between K_s and the water retention parameters:

$$K_s \propto (\theta_s - \theta_r) \left[\int_0^\infty rg(r) dr \right]^2 \quad [6]$$

Substituting Eq. [3] into Eq. [6] yields

$$K_s \propto \frac{\theta_e^{b+2}}{|\psi_m|^2} \exp(\sigma^2) \quad [7]$$

where b is a constant. Mualem (1976) demonstrated that $b = 0.5$ was the best value for 45 soil data sets. Moreover, because σ showed no correlation with $\log K_s$ (Fig. 13e) in this study, we can assume that σ was constant for all samples. Therefore, Eq. [7] becomes

$$K_s = B \frac{\theta_e^{2.5}}{|\psi_m|^2} \quad [8]$$

where B is a constant. Taking the logarithm of Eq. [8] yields

$$\log K_s = \log B + 2.5 \log \theta_e - 2 \log |\psi_m| \quad [9]$$

Equation [9] evaluates the effect of $\log \theta_e$ and $-\log |\psi_m|$ on $\log K_s$.

To determine the value of the constant B in Eq. [9], we minimized the residual sum of squares calculated using the observed and predicted $\log K_s$ of all Tanakami granite and Rokko granite samples. This procedure yielded the value $B = 232 \text{ cm}^3 \text{ s}^{-1}$. Using this constant value, we show in Fig. 14 the zones delineated by the equi- K_s lines theoretically predicted by Eq. [9] with $K_s = 10^{-6.5}$ to $10^{-1.5} \text{ cm}^3 \text{ s}^{-1}$. Each zone is filled with the same color

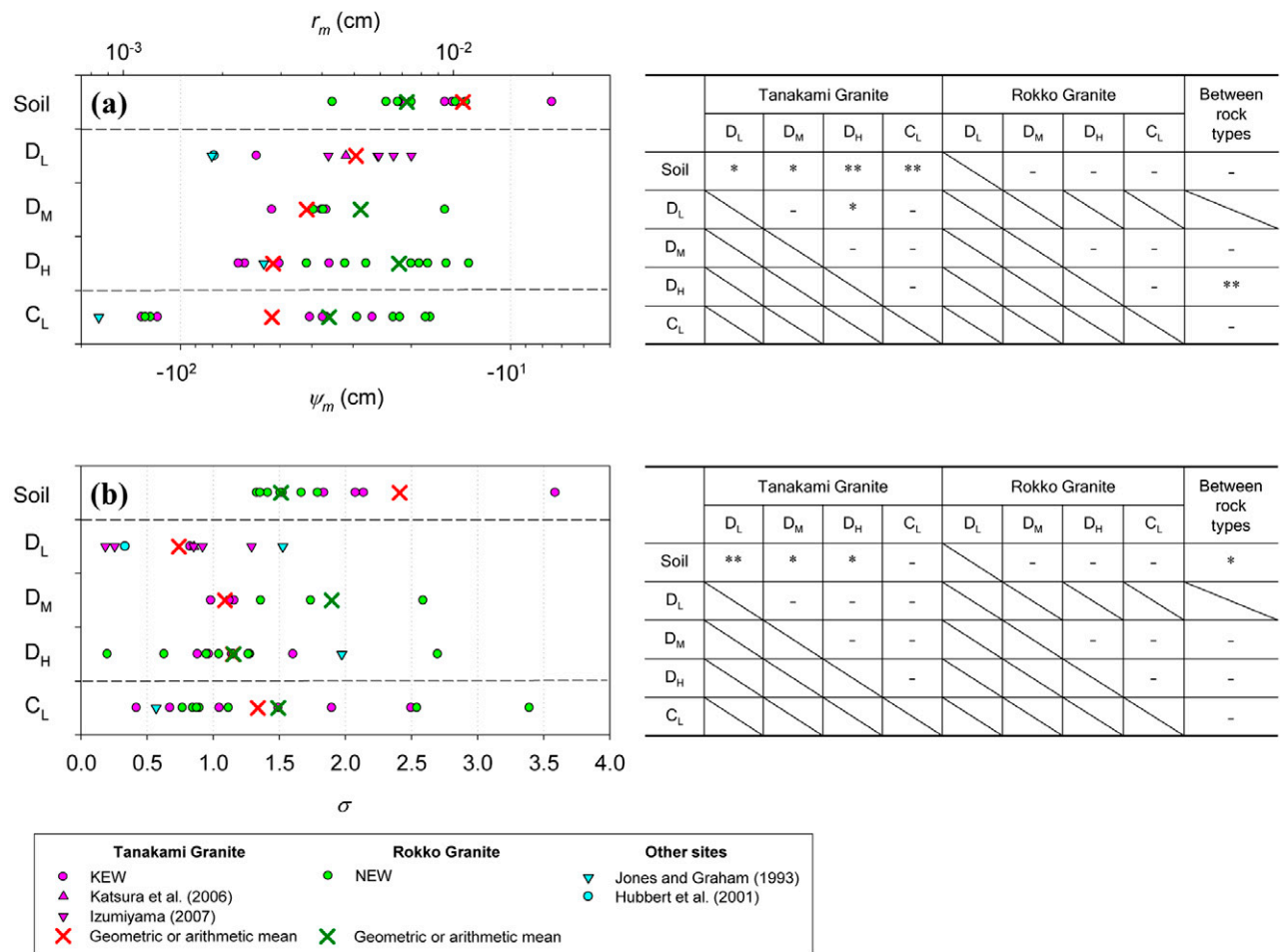


FIG. 11. Relationship between weathering class (see Table 1 for weathering class definitions) and (a) pressure head when the effective saturation (S_e) is 0.5 (ψ_m) or medium pore radius (r_m) and (b) the parameter characterizing the width of the pore size distribution (σ) at the Kiryu Experimental Watershed (KEW), the Nishi'otafuku-Yama Experimental Watershed (NEW), and other sites. The tables to the right show the significance level of the difference between the mean values of the corresponding categories: **significant at $P < 0.01$, *significant at $P < 0.05$, dash signifies not significant.

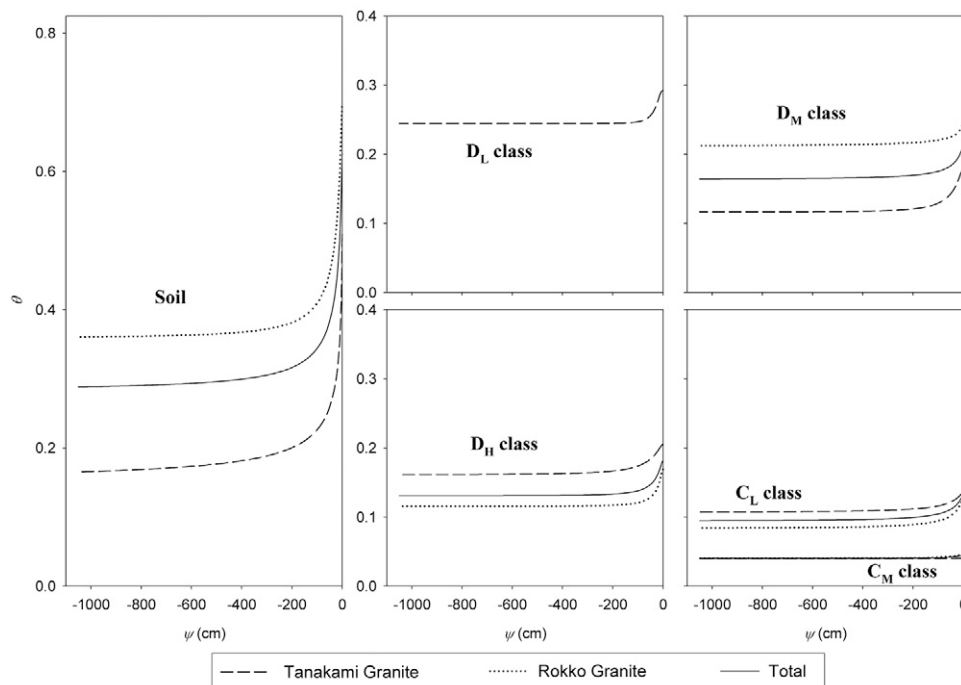


FIG. 12. Mean water retention curve for each weathering class (see Table 1 for weathering class definitions).

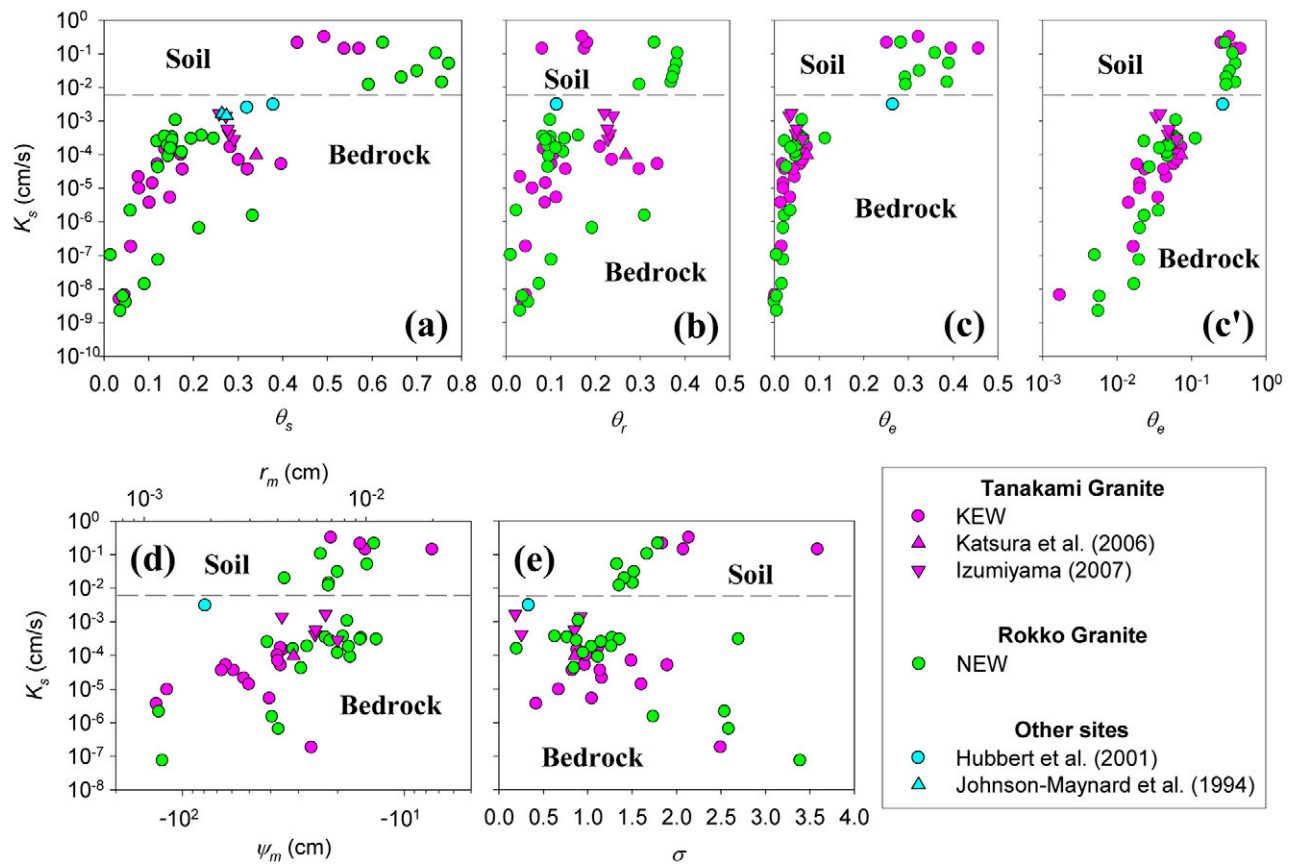


FIG. 13. Relationship between core-scale saturated hydraulic conductivity (K_s) and (a) saturated volumetric water content (θ_s), (b) residual volumetric water content (θ_r), (c) effective porosity (θ_e), (c') θ_e plotted on a logarithmic scale, (d) pressure head when the effective saturation (S_e) is 0.5 (ψ_m) or medium pore radius (r_m), and (e) the parameter characterizing the width of the pore size distribution (σ) at the Kiryu Experimental Watershed (KEW), the Nishi'otafuku-Yama Experimental Watershed (NEW), and other sites.

as the plot with the same K_s range. The plots for Tanakami granite and Rokko granite generally fell on or near the zone of the same color, suggesting that Eq. [9] successfully reproduced the observed K_s value of both the bedrock and soil samples. Thus, the model developed by Mualem (1976) can also be applied to core-scale bedrock hydraulic properties.

Changes in Pore Structures with Weathering Inferred from Core-Scale Hydraulic Properties

The water retention curve of the C_M -class bedrock exhibited little change in θ across a wide range of ψ (Fig. 12), which resulted in the extremely small θ_e values (Fig. 10c). This suggests that the pores in the C_M -class bedrock were almost all micropores, which hardly contributed to water flow, as reflected in the extremely low core-scale K_s values (Fig. 9). As weathering processes progressed and C_M -class bedrock was weathered into C_L class, the number of micropores, which did not release water at $\psi = -200$ cm, increased and macropores, which held water in the wet range where ψ was greater than -200 cm, appeared (Table 3). The increase in micropores led to the increase in θ_r , and the production of macropores contributed to the substantial increase in θ_e (Fig. 10). The latter effect consequently elevated core-scale K_s considerably (Fig. 9). A clear distinction between C_M - and C_L -class cores is that the former remained unweathered inside the rock body, while the latter were completely weathered (Table 1). Because this study used core samples without apparent

fractures, we can assume that all of the C_M -class samples represented fresh, almost unweathered bedrock. The early stage of weathering involves physical weathering processes, such as unloading. This can weaken the chemical bonding force between the rock particles (Japanese Society of Soil Mechanics and Foundation Engineering, 1985), creating some macropores in the C_L -class bedrock.

The water retention curves of the C_L - to D_L -class bedrock were nearly parallel to each other, showing that θ gradually changed with ψ in the wet range, but changed little in the dry range (Fig. 12). From these trends, we can infer that the C_L - to D_L -class bedrock contained relatively distinguishable macropores and micropores. The parallelism of the curves suggests that the increase in weathering from the C_L to D_H , D_M , and D_L classes increases the number of micropores greatly but does not increase the numbers of macropores (Table 3). The progress of weathering from C_L into D_L classes involves the alternation of minerals with little resistance to weathering, such as feldspar, into clay. Other minerals, however, such as quartz, retain the original rock fabric and structure (Table 1). Alteration of minerals into clay involves an increase in the interlayer water and water of crystallization (Japanese Society of Soil Mechanics and Foundation Engineering, 1985), which agrees with the observed increase in θ_r (i.e., micropores) with increasing weathering degree. As with θ_e (Fig. 10c), the change in the core-scale K_s with increasing weathering degree was generally not significant (Fig. 9). For C_L - to D_L -class bedrock,

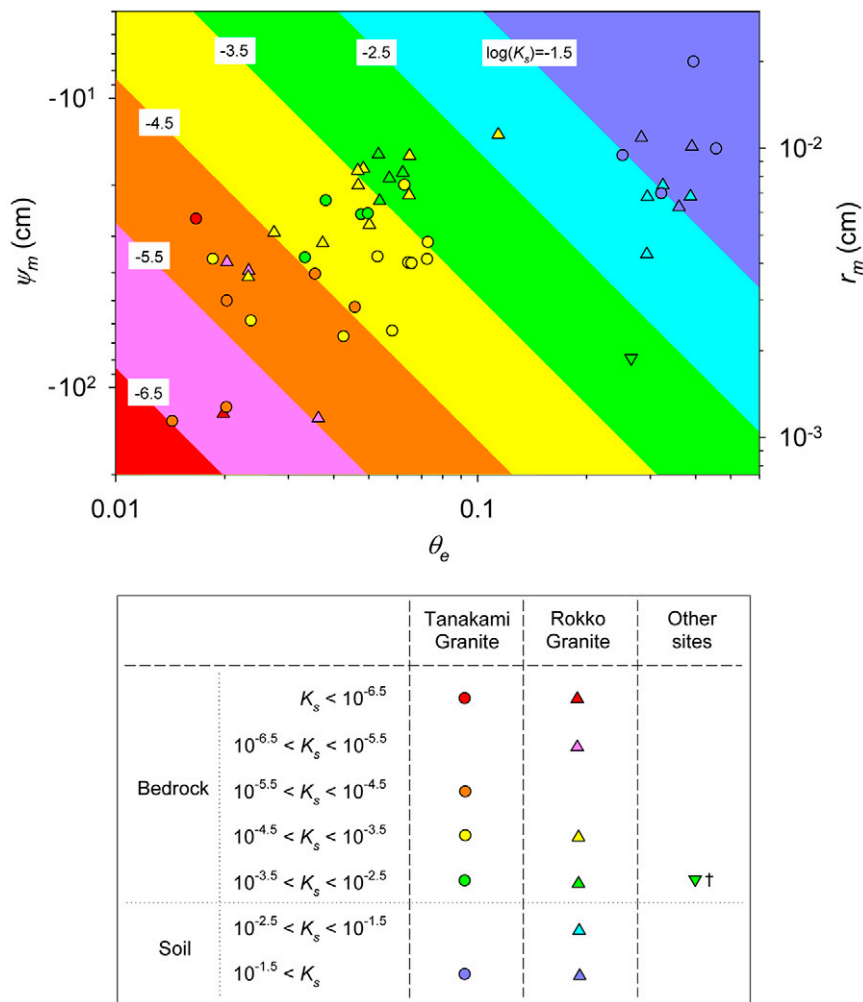


FIG. 14. Relationship between effective porosity (θ_e) and pressure head when the effective saturation (S_e) is 0.5 (ψ_m) or medium pore radius (r_m) on a logarithmic scale. The plots were classified according to the core-scale saturated hydraulic conductivity (K_s). The lines represent the theoretical relationship given by Eq. [9] with $K_s = 10^{-6.5}$ to $10^{-1.5}$ cm s⁻¹. The zones separated by these lines are filled with the same color as the plots with the same K_s range. † Data reported by Hubbert et al. (2001).

θ_e and ψ_m , rather than the weathering class, could be a better indicator for the core-scale K_s (Fig. 14).

As the D_L-class bedrock was weathered into soil, θ_e dramatically increased (Fig. 10c). The water retention curves (Fig. 12) of the soil displayed a much steeper change in θ than did those of the D_L-class bedrock in the wet range. Hence, most of the increase in θ_e from the D_L-class bedrock to the soil is attributable to the production of macropores, which held water in the wet range (Table 3). The production of many macropores in the soil also contributed to a jump in core-scale K_s compared with the bedrock (Fig. 9). The collapse of rock fabric and structure, together with biological effects such as plant root invasion and earthworm activity, probably caused the inferred changes in the pore structures.

Comparison with the Core-Scale Hydraulic Properties of Weathered Granitic Bedrock at Other Sites

A few previous studies measured the K_s (Johnson-Maynard et al., 1994), WRC (Jones and Graham, 1993), and both K_s and WRC (Hubbert et al., 2001) of weathered granitic bedrock samples

in the laboratory. We determined the weathering class of the samples used in these studies based on their characteristics, and included their data points in Fig. 9 to 11 and 13.

Figure 9 demonstrates that the core-scale K_s of D_L-class bedrock showed no significant difference between our sites (i.e., Tanakami granite and Rokko granite) and the other sites. One data point for the D_H-class sample of the other site lies in a larger K_s range than in our sites, but it is within the range of K_s for the C_L- to D_L-class bedrock group described above. Likewise, no significant difference in the θ_s (Fig. 10a), θ_r (Fig. 10b), and σ (Fig. 11b) values of the C_L- to D_L-class bedrock group was found between our sites and the other sites. The θ_s , θ_r , and θ_e of one C_M-class sample in the other site lie well within the data range for our sites (Fig. 10). Furthermore, the data points in the other sites generally follow the correlations between K_s and the water retention parameters observed in our sites (Fig. 13). A clear difference between our sites and the other sites is that the θ_e (Fig. 10c) and ψ_m (Fig. 11a) values of D_L-class bedrock in the other sites were significantly larger and smaller, respectively ($P < 0.01$). One of the corresponding data points ($\theta_e = 0.265$, $\psi_m = -79.0$ cm, and $K_s = 3.1 \times 10^{-3}$ cm s⁻¹), however, reported by Hubbert et al. (2001) with a complete set of these parameters, was plotted in Fig. 14 in the zone of the same color, suggesting that it could be explained by Eq. [9] with a value of B equivalent to that at our sites. The limited availability of data from the other sites precludes more detailed analysis and discussion here. Nonetheless, these results, combined with the fact that no significant differences were observed between Tanakami granite and Rokko granite, suggest that it may be possible to treat the hydraulic properties of similarly weathered granitic bedrock equally.

Modes of In Situ Water Flow through Weathered Bedrock

From the discussion above, the weathered bedrock can be classified into two groups from the viewpoint of core-scale hydraulic properties: C_M-class bedrock and C_L- to D_L-class bedrock. While the core-scale K_s values of C_M-class bedrock were much smaller than the in situ values, the C_L- to D_L-class bedrock exhibited similar K_s values between the core-scale and in

TABLE 3. Change in the number of micropores and macropores in the matrix with increasing weathering degree.

Weathering class†	Micropores‡	Macropores§
From C _M - to C _L -class bedrock	+	+
From C _L - to D _L -class bedrock	+	±
From D _L -class bedrock to soil	+	++

† See Table 1 for weathering class definitions.

‡ Pores that do not release water at a pressure head (ψ) of -200 cm; + signifies a moderate increase.

§ Pores that hold water in the wet range where $\psi > -200$ cm; ± signifies no significant increase, ++ signifies a large increase.

situ measurements (Fig. 4 and 5). The core shape was generally II to IV for C_M -class bedrock, which generally showed clearly distinguishable matrix and fractures, and IV to VII for C_L - to D_L -class bedrock, which showed a dominance of matrix rather than apparent fractures (Fig. 1–3). The progress of weathering from C_L into D_L classes involved unloading and alteration of minerals into clay, and we often observed fractures clogged with clayey materials. These increasing effects of weathering could account for the core shape observed for C_L - to D_L -class bedrock. From the K_s values and the core shape, it can be inferred that in situ water flow through C_L - to D_L -class bedrock is well represented by matrix flow, as captured by the core-scale measurements. In other words, in situ water flow through C_L - to D_L -class bedrock can probably be described by Darcy's law. In situ water flow through C_M -class bedrock, on the other hand, is restricted to fractures, with almost no contribution of the matrix to the total flow. Thus, core-scale measurements cannot represent in situ water flow through C_M -class bedrock. During the drilling of boreholes through the C_M -class bedrock, we observed substantial water leakage, which could not be explained by the core-scale K_s values. This may also support our inference.

The in situ K_s values measured at other sites may additionally support our reasoning. We found a few previous studies (Graham et al., 1997; Frazier et al., 2002; Tromp-van Meerveld et al., 2007) that can be regarded as essentially evaluating the in situ K_s values of D_L -class granitic bedrock. Figure 9, which includes the data points for these studies, shows no significant difference between in situ and core-scale values measured in our sites and the other sites ($P > 0.05$). This implies that the in situ water flow through D_L -class bedrock can be characterized by the core-scale K_s values. Again, the limited data available provide insufficient support for the discussion here. They do imply, however, that the results of this study may have wide applicability.

Summary and Conclusions

We measured the K_s and WRCs of weathered granitic bedrock core samples collected from five boreholes (total depth, 12–38 m) in two headwater catchments (Tanakami granite and Rokko granite) using accepted soil physics methods. The core samples showed various degrees of weathering, classified into weathering classes C_M (weakly weathered) to D_L (highly weathered). Essentially no significant differences in the obtained core-scale hydraulic properties were found between the Tanakami granite and Rokko granite. We concluded that the bedrock could be classified into two groups from the viewpoint of the hydraulic properties: C_M -class bedrock and C_L - to D_L -class bedrock.

The C_M -class bedrock is characterized by extremely low core-scale K_s , with a geometric mean of approximately 10^{-8} cm s $^{-1}$, and a water retention curve that shows little change in θ across the entire range of ψ greater than -1000 cm. These core-scale hydraulic properties suggest that almost all of the C_M -class bedrock pores were micropores, which did not significantly contribute to water flow. Flow through fractures is considered to be dominant to matrix flow in these cases.

The C_L - to D_L -class bedrock had a geometric mean core-scale K_s ranging from 10^{-5} to 10^{-3} cm s $^{-1}$. A gradual change in θ with ψ was found in the wet range (ψ greater than -200 cm), while θ changed little in the dry range (ψ less than -200 cm). The water retention curves were nearly parallel to each other;

although θ_r increased with the weathering degree, θ_c did not increase substantially. This suggests that the number of micropores does increase with weathering degree, but that the volume of macropores, which contribute to water flow, does not increase greatly. Flow through these bedrock layers can probably be represented by matrix flow, as described by Darcy's law, rather than by preferential flow through fractures. While K_s cannot be predicted from the weathering class, it is instead well explained by the parameters characterizing the water retention curve.

We successfully measured the hydraulic properties of variously weathered granitic bedrock core samples and clarified their changes with increases in weathering. The fact that there were no essential differences between Tanakami granite and Rokko granite, and that the data from other sites followed the trends observed for our sites, suggests the possibility that the findings of this study will hold for various types of weathered granite. The hydraulic properties that we presented are fundamental to the analysis of water flow through weathered bedrock. Future work should further investigate the effects of fractures and analyze water flow processes through weathered bedrock. Such studies will contribute greatly to our understanding of the roles of weathered bedrock in runoff generation, stream water chemistry determination, and landslide occurrence in headwater catchments.

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